

Ian Lewis

Software Engineer

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TECHNICAL EXPERIENCE

Mozilla Open Source Initiative, Open-Source Software Contributor 09/2026 – Present

- Contribute patches across three Firefox components on bugs filed and prioritized by Mozilla engineers, investigating defects against an unfamiliar multi-million-line codebase and documenting proposed fixes for module owner review before writing code, then taking each change through local builds, mach lint and test runs, and review-gated Phabricator sign-off.
- Updated Firefox's Python-based Telemetry test harness, removing deprecated xUnit logging configuration and eliminating unused XML report generation from automated test runs.
- Eliminated a probabilistic ID collision in Firefox's tab group creation: the prior timestamp-plus-random scheme drew from 101 values per millisecond, reaching ~85% collision odds when extensions bulk-created 20 groups in a single millisecond. Replaced it with a SessionStore-persisted monotonic counter that reduced the collision rate to zero by construction, while preserving backward compatibility across session store files, DOM element IDs, and the WebExtensions ID mapping layer so existing user sessions required no migration.
- Developed a refactor of Firefox's multi-selection tab counting API, consolidating two overlapping count getters into one and extracting a private unsorted helper so the count matches selectedTabs.length structurally rather than by convention, migrating 8 production call sites and the affected test suite.

Trickl, Co-Creator, Software Engineer 02/2026 – 08/2026

- Designed a service-oriented telemetry ingestion pipeline using Vector, routing real-time data to VictoriaMetrics for Prometheus-compatible querying while independently archiving the unmodified stream to S3 for long-term retention.
- Built an OpenTelemetry cardinality management system analyzing each label's contribution to total series count, persisting recommendations to PostgreSQL and applying approved configuration changes via hot reload with zero downtime.
- Provisioned the full AWS architecture as code using CDK and TypeScript, deploying containerized services across ECS with an Application Load Balancer, Auto Scaling Groups, and a custom VPC with IAM-controlled service access, Cognito, Secrets Manager, and CloudWatch.
- Collaborated in a four-person distributed team through design reviews, pair programming, and technical decision making, building the Dockerized development environment, CLI, and React visualization layer alongside the backend services, and authored the public case study documenting the system's architecture and engineering tradeoffs.

Independent Projects & Client-Style Applications, Software Engineer 09/2023 – 03/2026

- The Dugout 🌐 (**Python, FastAPI, React, Vite, PostgreSQL, Pinecone, OpenAI API, Vercel, Render**). Built a retrieval-augmented Q&A service in Python: ingested and chunked source material, generated embeddings for vector storage, and orchestrated semantic search against user queries with a token-efficient generation model behind a FastAPI backend.
- LineupCFB 🌐 (**React, Supabase/PostgreSQL, Node.js, Express, Jest, Playwright, GitHub Actions, Capacitor**). Full-stack fantasy college football platform shipped to the Apple App Store and maintained through a live season, with server-side logic split between a Node cron process and PostgreSQL functions and RLS policies.
 - Built and maintain a 443-case automated test suite covering scoring, captain assignment, waiver rules, and scheduling, structured so the core logic can be tested without a live database. Added a test that fails whenever the league's rule constants fall out of sync between the frontend, backend, and database, and run end-to-end browser tests against the production build on desktop and mobile in CI.
- RequestStore (**Node.js, Express, React, TypeScript, MongoDB**). Self-hosted RequestBin-style tool for capturing and inspecting incoming webhooks and HTTP requests during integration debugging. Deployed to a hand-built AWS VPC across public and private subnets and multiple availability zones with internet and NAT gateways, and Nginx configured as a reverse proxy on Linux hosts.

SKILLS

Languages: JavaScript, TypeScript, Python, Ruby, SQL, HTML/CSS

Testing & Developer Tools: Jest, Playwright, node:test, React Testing Library, end-to-end and cross-viewport testing, Postman

CI/CD Tooling: GitHub Actions, Git, Docker, test pipelines, Infrastructure as Code (AWS CDK)

Systems & Cloud: Linux, AWS, CDK, ECS, Cloud Networking, Nginx, Vercel, Render

Backend & Data: Node.js, Express, FastAPI, REST APIs, PostgreSQL, MongoDB, Redis, Supabase, S3

Observability: Prometheus, OpenTelemetry, Grafana, VictoriaMetrics, Mimir, Honeycomb

AI Engineering: OpenAI API, Claude Code, Codex, Pinecone

ADDITIONAL EXPERIENCE

Educator & Administrator, UK & US School Systems 08/2013 – 09/2023 | UK, US

- Mentored early-career teachers through structured observation and feedback cycles, and coordinated across departments to deliver curriculum, staff development, and budget planning against fixed academic deadlines.
- Designed a maths curriculum adopted county-wide across multiple schools, documenting it for delivery by teachers I did not manage and iterating on it from classroom feedback; math results improved 32% year over year.

EDUCATION

MBA, Northeastern State University
Graduated with Honors.

09/2021 – 12/2022

BS in Education, Kingston University
Graduated with Honors.

09/2010 – 07/2013 | London, England